

Ch 4 Applications of Derivatives

One of the most important applications of the derivative is its use as a tool for finding the optimal (best) solutions to problems. In this chapter we apply derivatives to find extreme values of functions, to determine and analyze the shapes of graphs, and to solve equations numerically. We also introduce the idea of recovering a function from its derivative. The key to many of these applications is the Mean Value Theorem, which connects the derivative and the average change of a function.

4.1 Extreme Values of Functions on Closed Intervals

This section shows how to locate and identify extreme (maximum or minimum) values of a function from its derivative. Once we can do this, we can solve a variety of optimization problems (see Section 4.5). The domains of the functions we consider are intervals or unions of separate intervals.

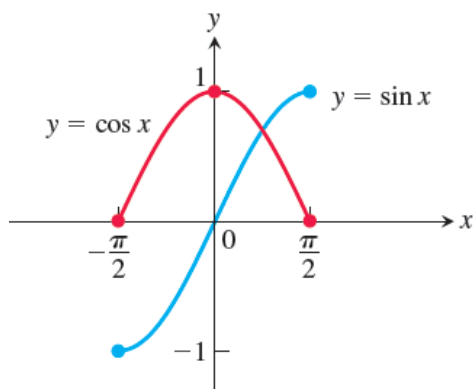
① Absolute (global) maximum and minimum

DEFINITIONS Let f be a function with domain D . Then f has an **absolute maximum** value on D at a point c if

$$f(x) \leq f(c) \text{ for all } x \text{ in } D$$

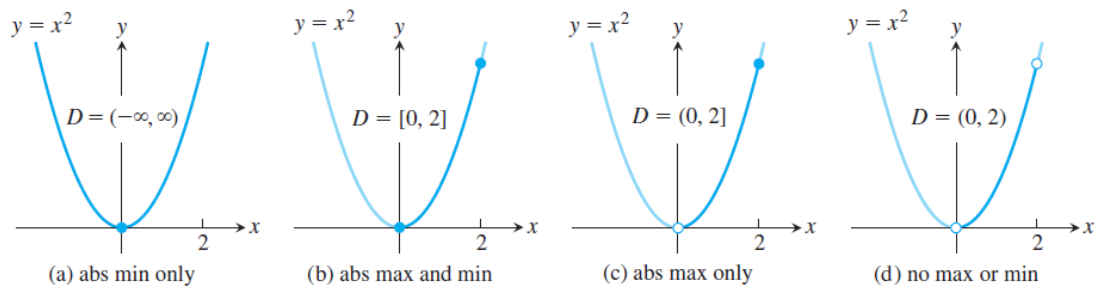
and an **absolute minimum** value on D at c if

$$f(x) \geq f(c) \text{ for all } x \text{ in } D.$$



Maximum and minimum values are called **extreme values** of the function f .

Example 1 The absolute extrema of the following functions on their domains can be seen in the following Figure. Each function has the same defining equation, $y = x^2$, but the domains vary.



Function rule	Domain D	Absolute extrema on D
(a) $y = x^2$	$(-\infty, \infty)$	No absolute maximum Absolute minimum of 0 at $x = 0$
(b) $y = x^2$	$[0, 2]$	Absolute maximum of 4 at $x = 2$ Absolute minimum of 0 at $x = 0$
(c) $y = x^2$	$(0, 2]$	Absolute maximum of 4 at $x = 2$ No absolute minimum
(d) $y = x^2$	$(0, 2)$	No absolute extrema

Note 1 Some of the functions in Example 1 do not have a maximum or a minimum value. Then a question arises: under what conditions must a function have a maximum or a minimum value?

THEOREM 1 (The Extreme Value Theorem)

If f is continuous on a closed interval $[a, b]$, then f attains both an absolute maximum value M and an absolute minimum value m in $[a, b]$. That is, there are numbers x_1 and x_2 in $[a, b]$ with

$$f(x_1) = m, f(x_2) = M,$$

and $m \leq f(x) \leq M$ for every other x in $[a, b]$.

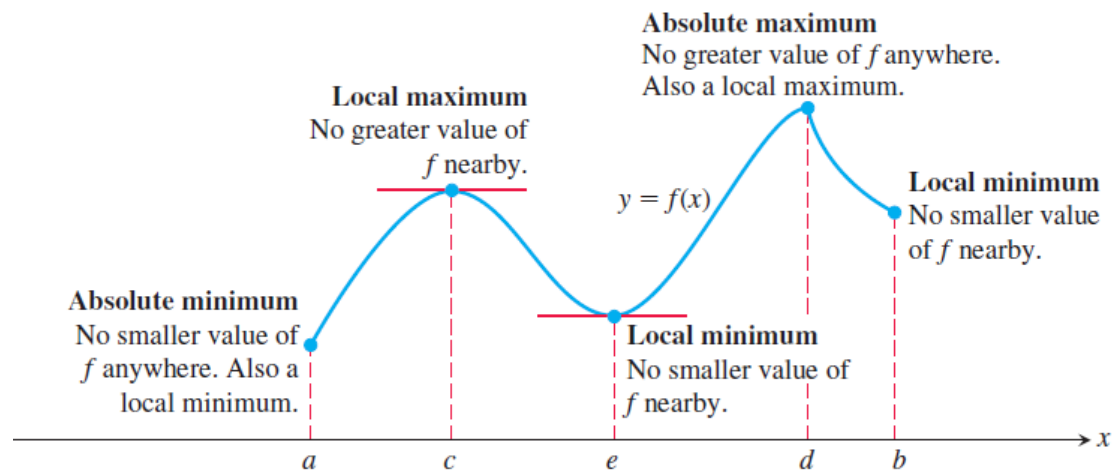
② Local (Relative) Extreme Values

DEFINITIONS A function f has a **local maximum** value at a point c within its domain D if $f(x) \leq f(c)$ for all $x \in D$ lying in some open interval containing c .

A function f has a **local minimum** value at a point c within its domain D if $f(x) \geq f(c)$ for all $x \in D$ lying in some open interval containing c .



If the domain of f is the closed interval $[a, b]$, then f has a local maximum at the endpoint $x = a$ if $f(x) \leq f(a)$ for all x in some half-open interval $[a, a + \delta)$, $\delta > 0$. Likewise, f has a local maximum at an interior point $x = c$ if $f(x) \leq f(c)$ for all x in some open interval $(c - \delta, c + \delta)$, $\delta > 0$, and a local maximum at the endpoint $x = b$ if $f(x) \leq f(b)$ for all x in some half-open interval $(b - \delta, b]$, $\delta > 0$.



③ Finding Extrema

THEOREM 2 (The First Derivative Theorem for Local Extreme Values)

If f has a local maximum or minimum value at an interior point c of its domain, and if f' is defined at c , then

$$f'(c) = 0.$$

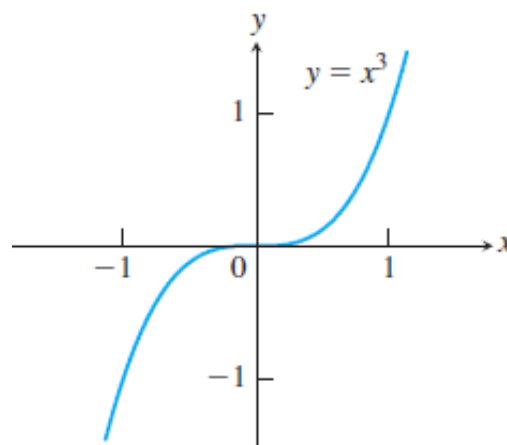
Note 2 $f(c)$ is an interior local maximum or minimum $\Rightarrow f'(c) = 0$ or $f'(c)$ is undefined.

Note 3 The only places where a function f can possibly have a local extreme value are

1. interior points where $f' = 0$,
2. interior points where f' is undefined,
3. endpoints of the domain of f .

Note 4 $f'(c) = 0 \not\Rightarrow f(c)$ is an interior local maximum or minimum.

For example, $f(x) = x^3$ on $x = 0$.



DEFINITION An interior point of the domain of a function f where f' is zero or undefined is a **critical point** of f .

Most problems that ask for extreme values call for finding the extrema of a continuous function on a closed and finite interval. Theorem 1 assures us that such values exist; Theorem 2 tells us that they are taken on only at critical points and endpoints.

◆ Finding the Absolute Extrema of a Continuous Function f on a Finite Closed Interval

1. Find all critical points of f on the interval.
2. Evaluate f at all critical points and endpoints.
3. $\max$ = the largest of these values, $\min$ = the smallest of these values.

Example 2 Find the absolute maximum and minimum values of

$$f(x) = x^2$$

on $[-2, 1]$.

Example 3 Find the absolute maximum and minimum values of

$$g(t) = 8t - t^4$$

on $[-2, 1]$.

Example 4 Find the absolute maximum and minimum values of $f(x) = x^{\frac{2}{3}}$ on the interval $[-2, 3]$.